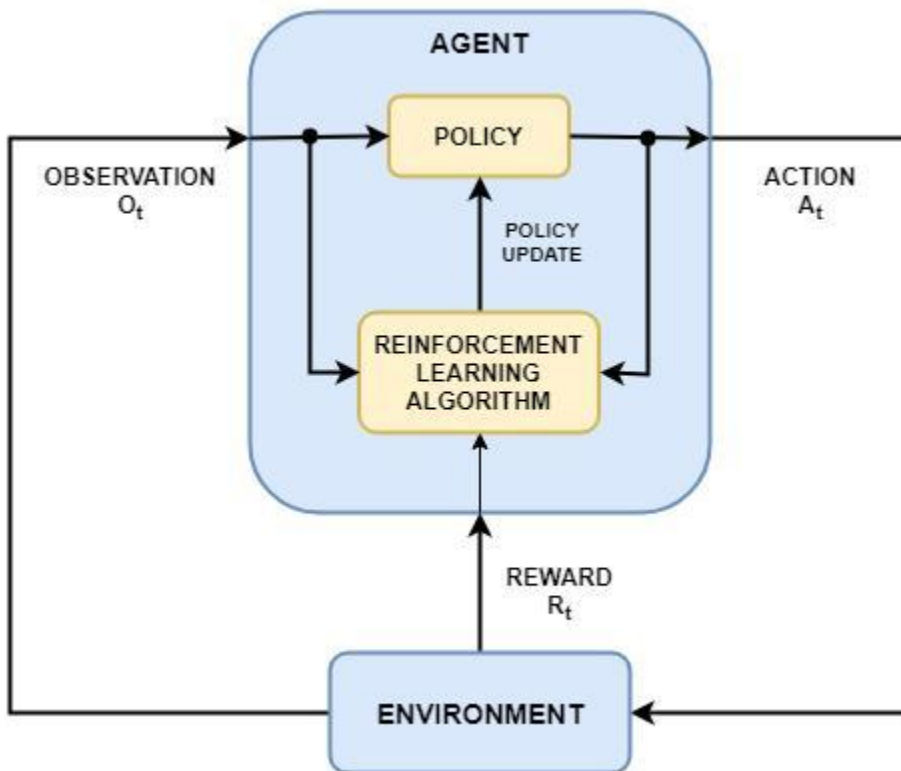
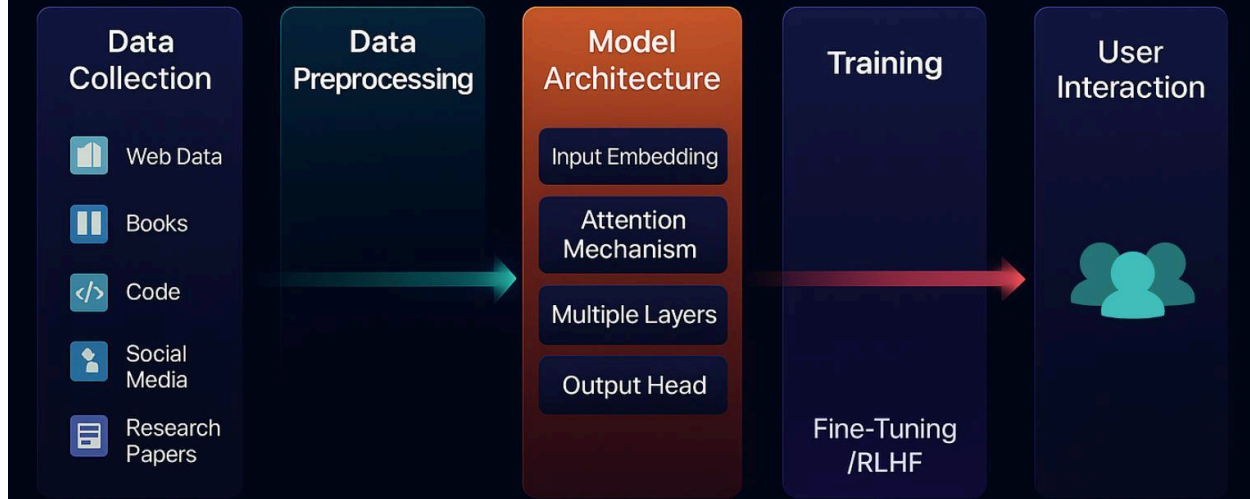
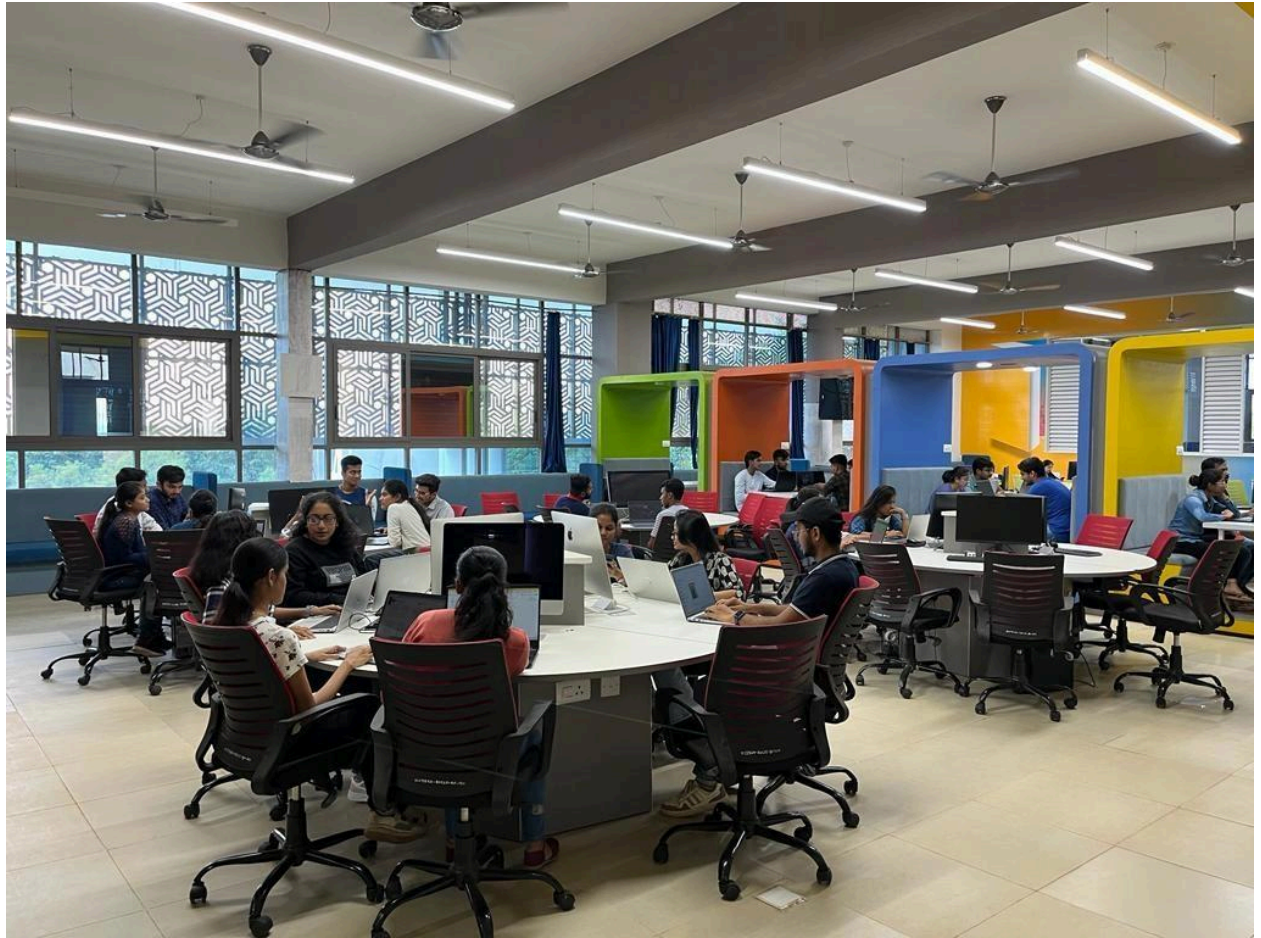


🧠 AI Research Engineer Roadmap (Deep Learning, LLMs, RL)

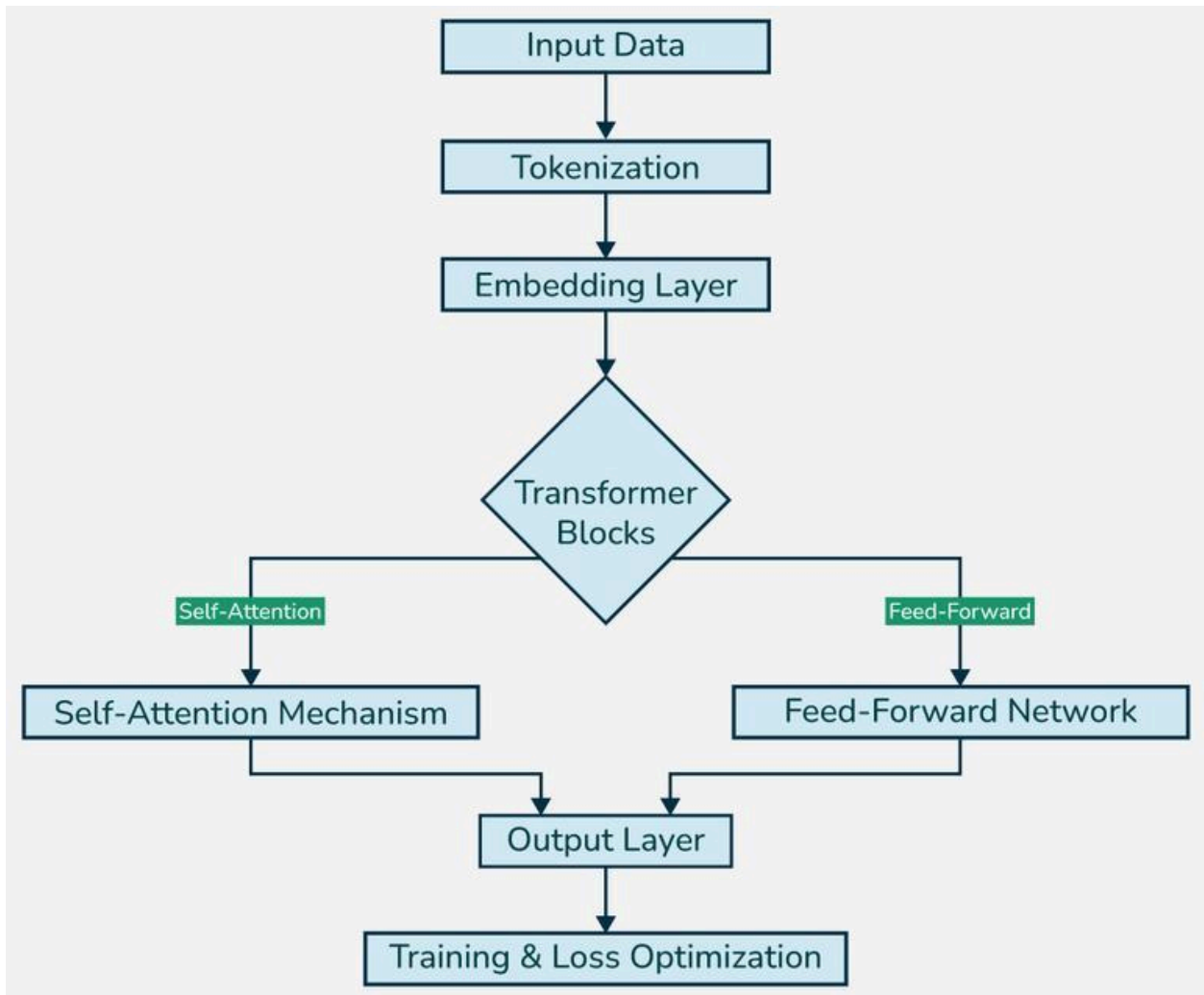


Large Language Model

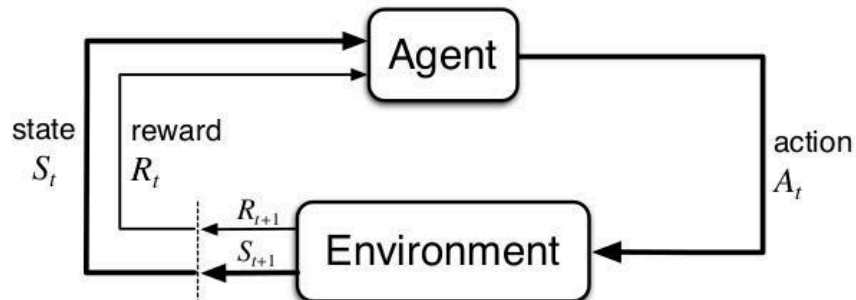








The Agent-Environment Interface



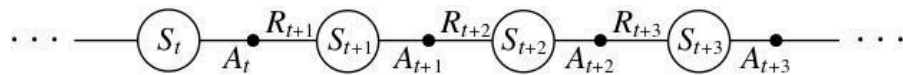
Agent and environment interact at discrete time steps: $t = 0, 1, 2, 3, \dots$

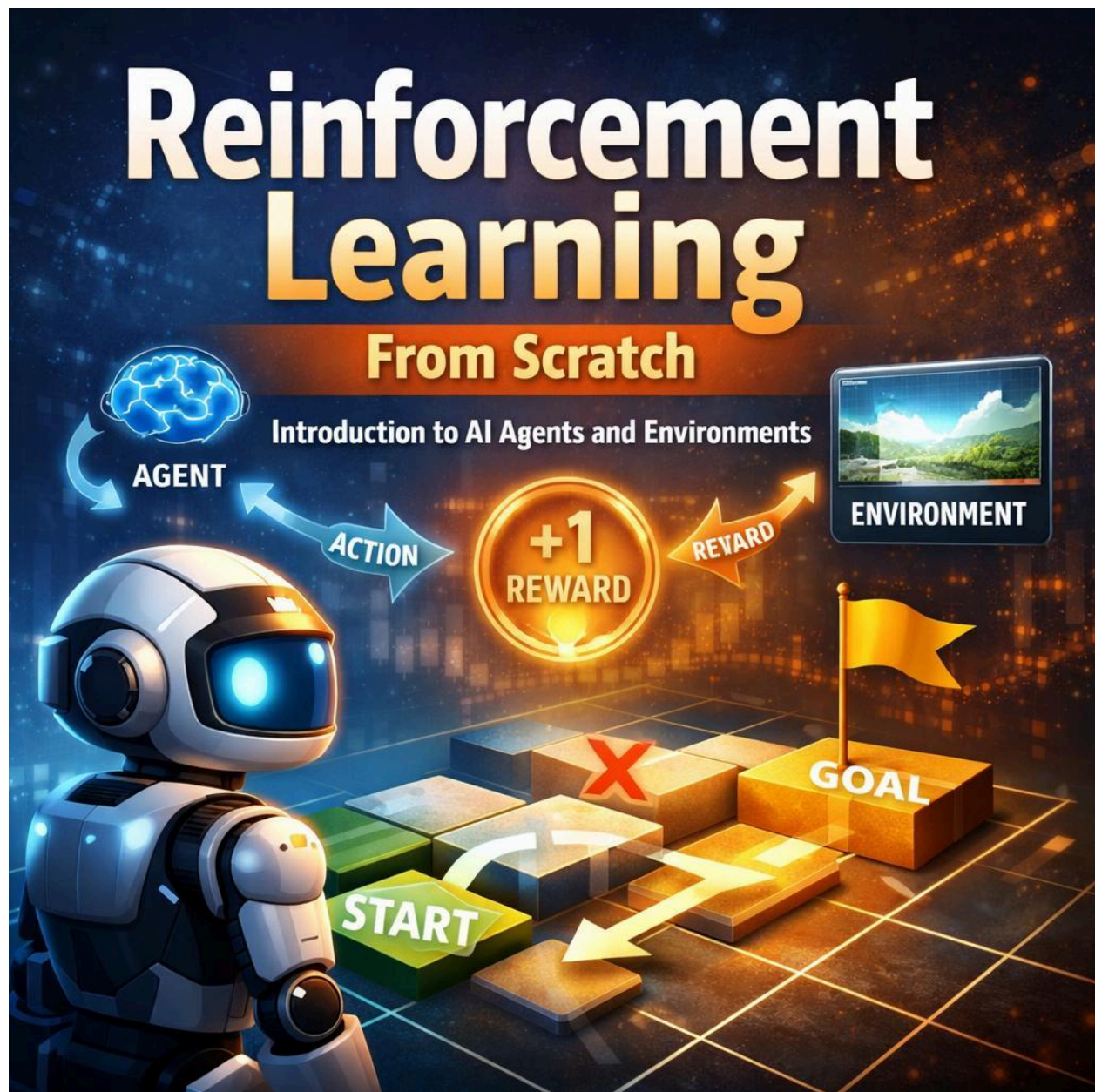
Agent observes state at step t : $S_t \in \mathcal{S}$

produces action at step t : $A_t \in \mathcal{A}(S_t)$

gets resulting reward: $R_{t+1} \in \mathcal{R} \subset \mathbb{R}$

and resulting next state: $S_{t+1} \in \mathcal{S}^+$





An AI Research Engineer develops new AI models, improves existing algorithms, publishes research, and builds advanced systems such as LLMs, multimodal AI, robotics AI, and autonomous agents.

Phase 1: Mathematics Foundation

Learn

- Linear Algebra
- Probability & Statistics
- Calculus
- Optimization
- Information Theory

Resources

- Mathematics for Machine Learning
 - Deep Learning
-

Phase 2: Python & Scientific Computing

Tools

`pip install numpy pandas scipy matplotlib seaborn jupyter`

Learn

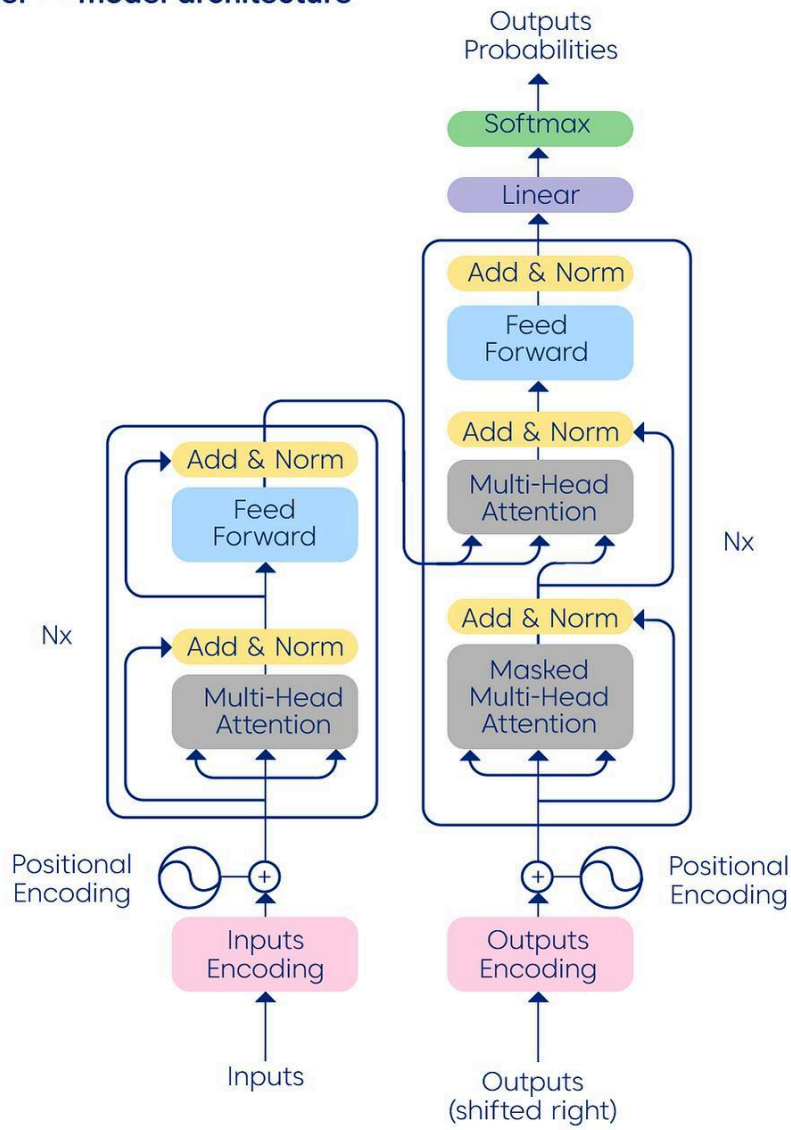
- NumPy
- Pandas
- SciPy
- Data Visualization
- Algorithm Design

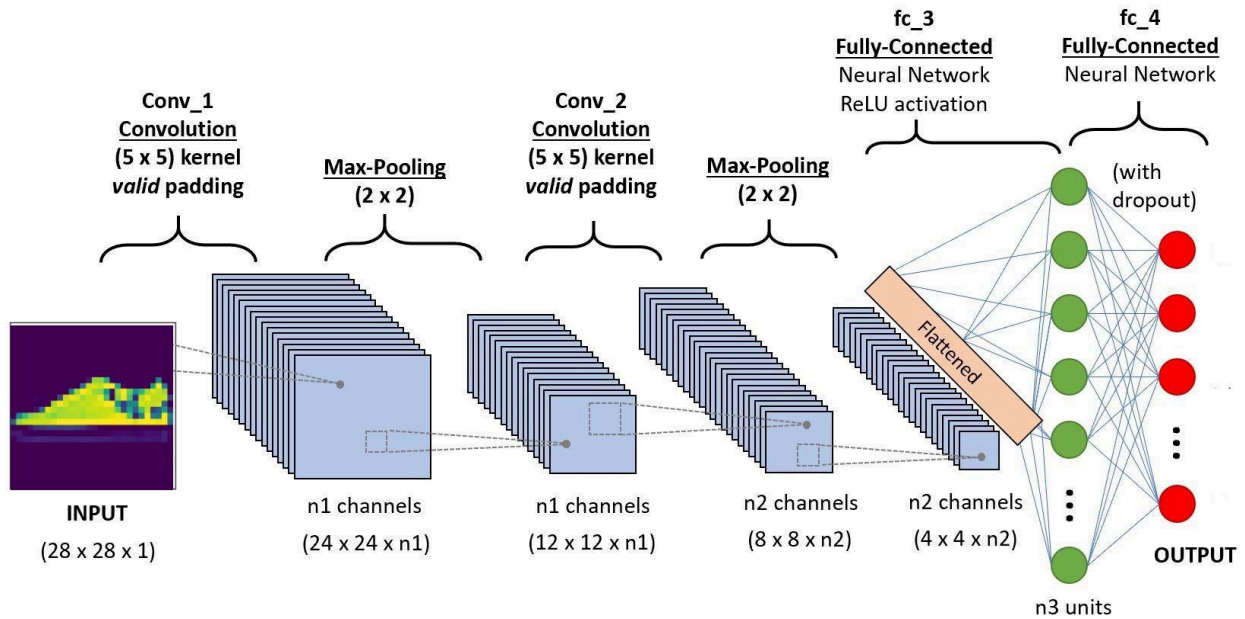
Projects:

- Data Analysis
 - Optimization Algorithms
 - Scientific Simulations
-

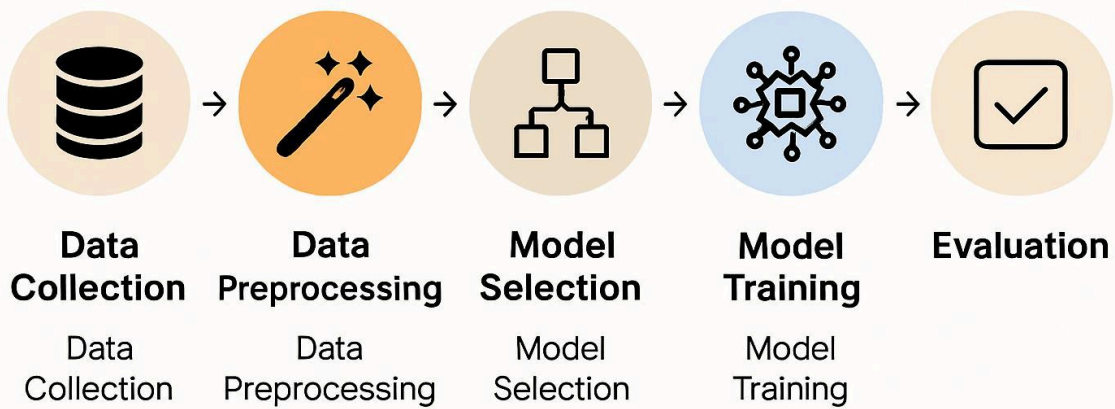
Phase 3: Deep Learning

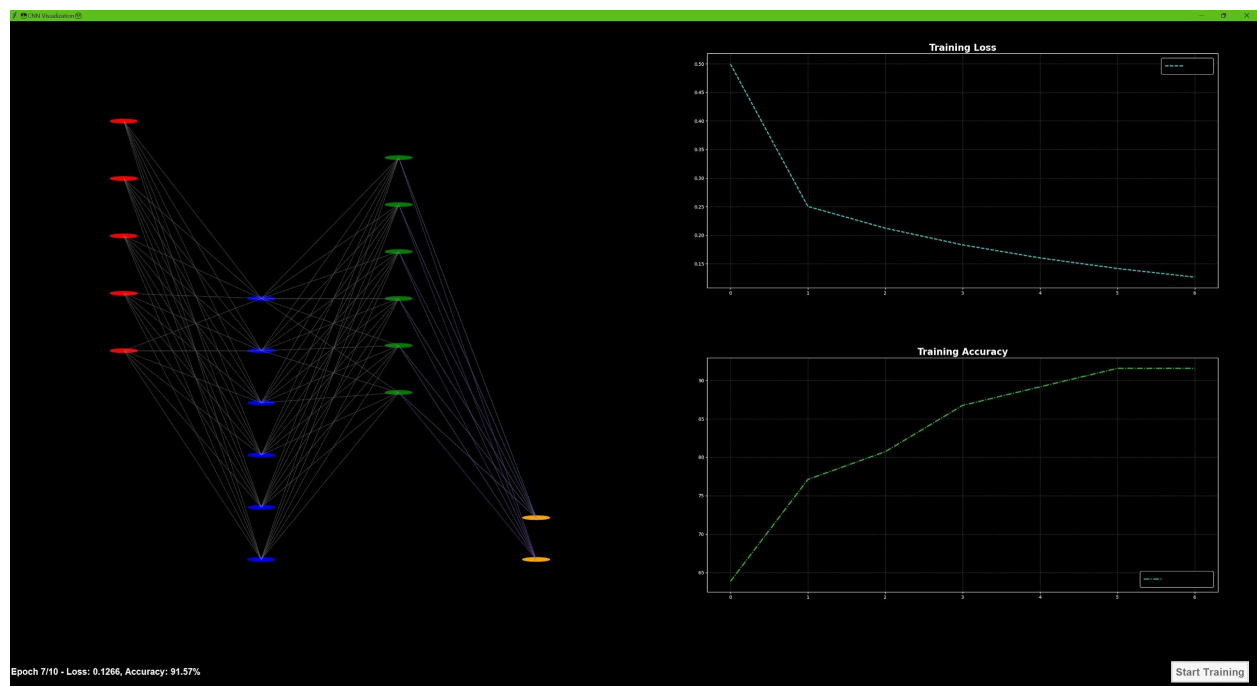
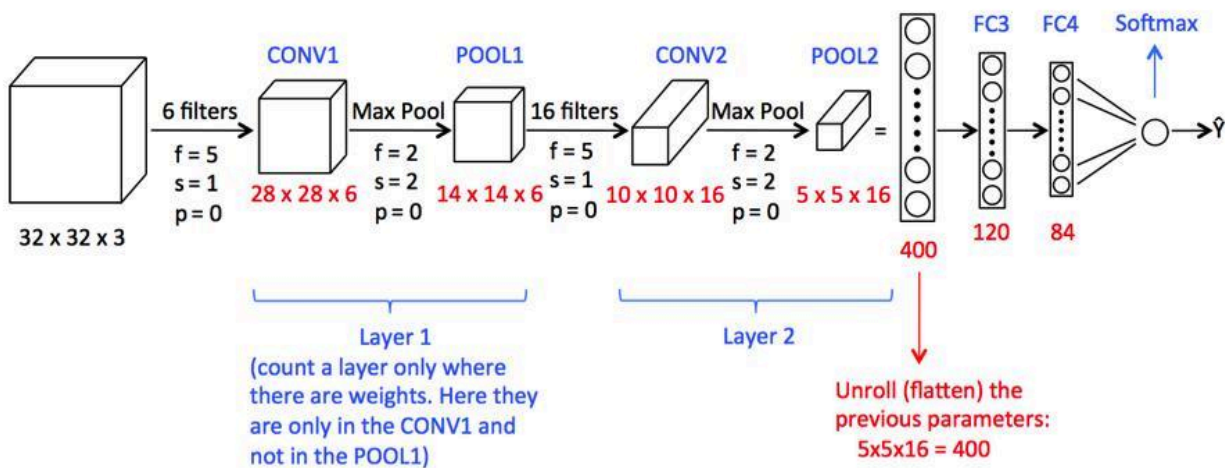
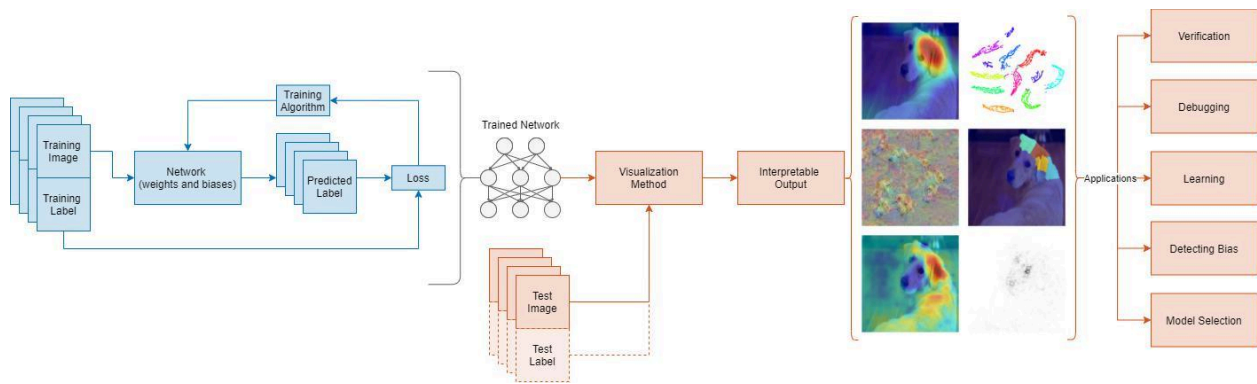
The Transformer — model architecture





TRAINING AN AI MODEL







Frameworks

- [PyTorch](#)
- [TensorFlow](#)

Learn

- ANN
- CNN
- RNN
- LSTM
- GRU
- Transformers
- Attention Mechanisms

Projects:

- Image Classification
- Face Recognition
- Object Detection
- Medical AI

Phase 4: Large Language Models (LLMs)

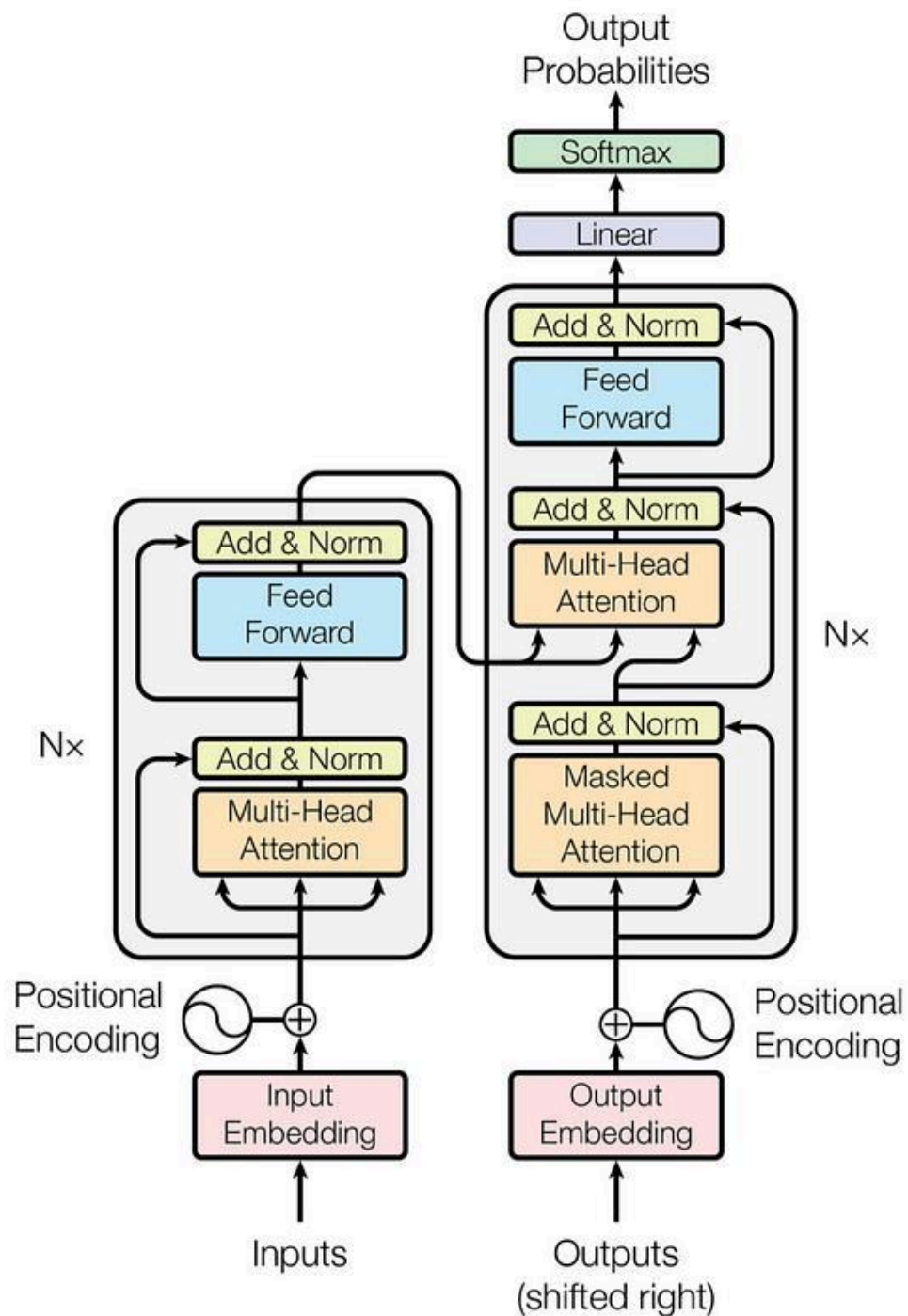
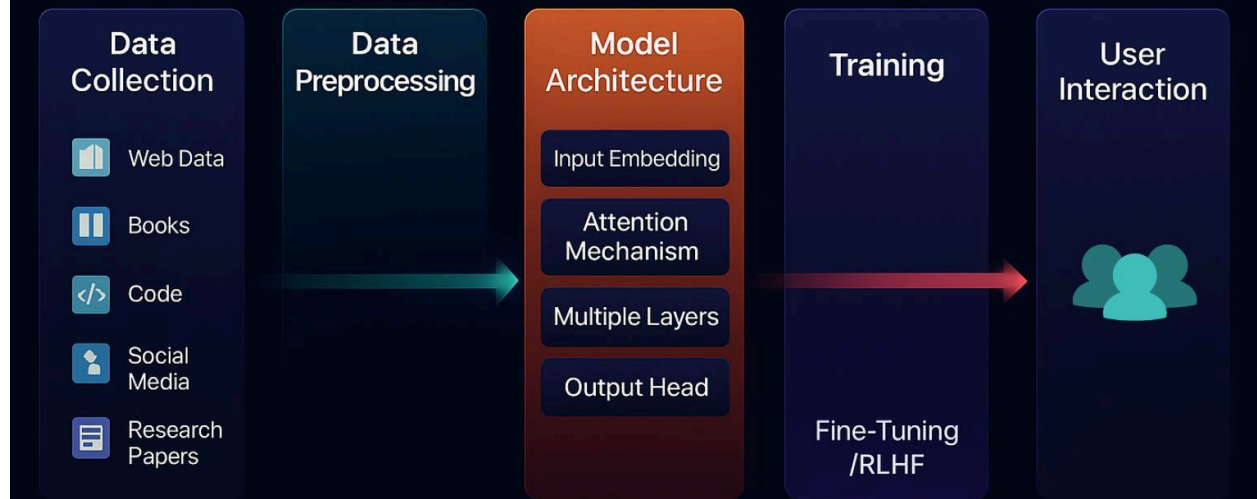
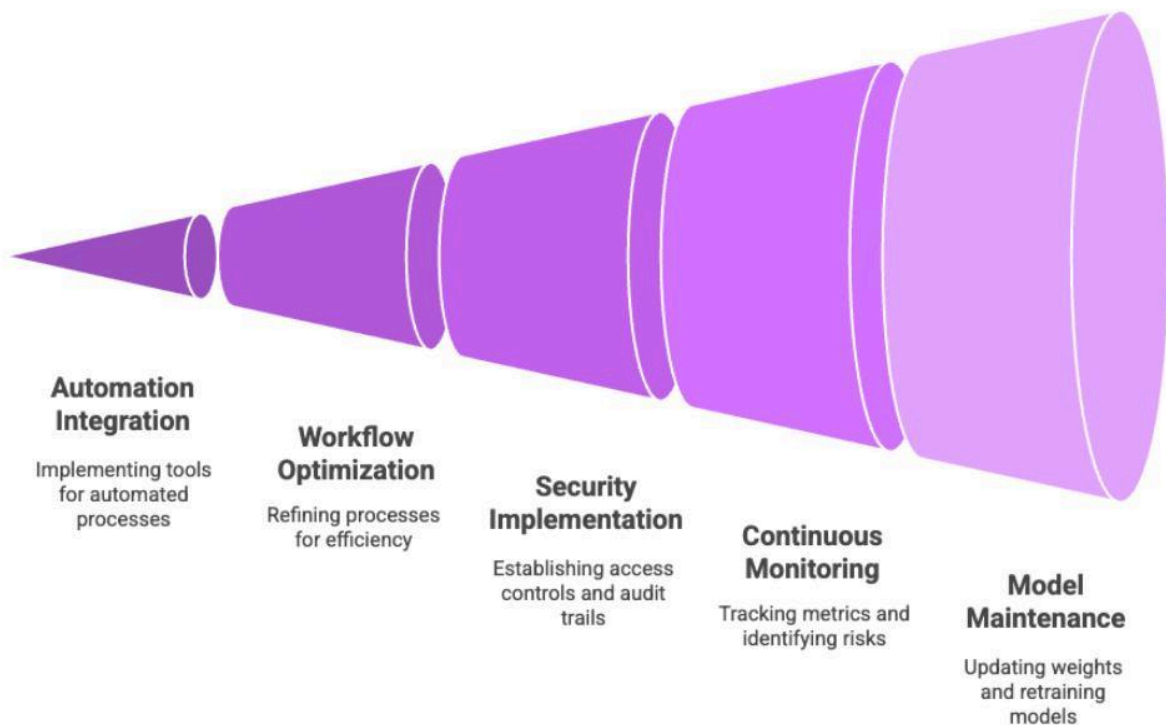


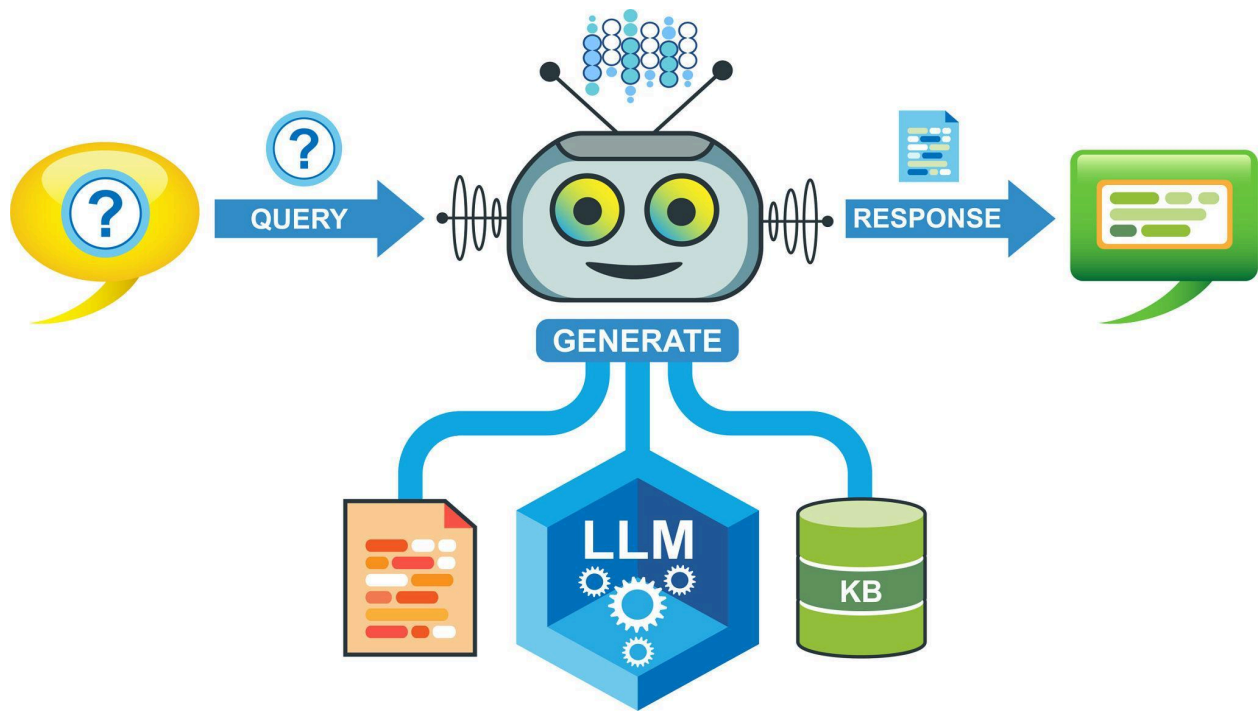
Figure 1: The Transformer - model architecture.

Large Language Model

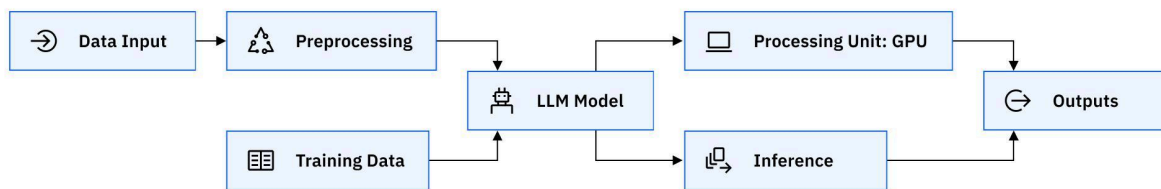


Streamlining LLM Training Pipeline

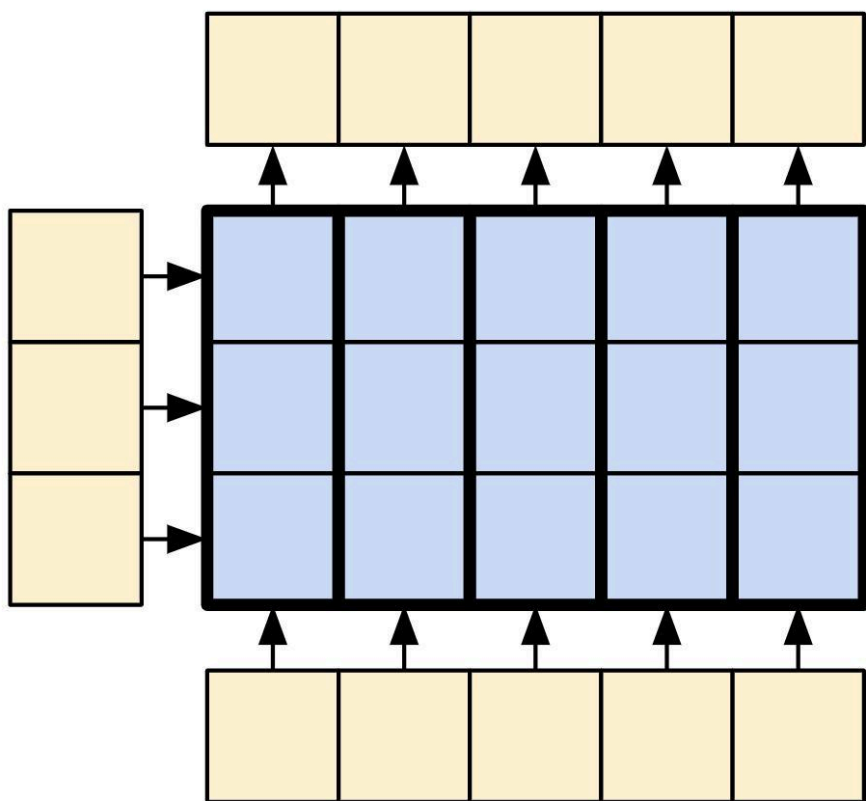




Traditional LLM training architecture

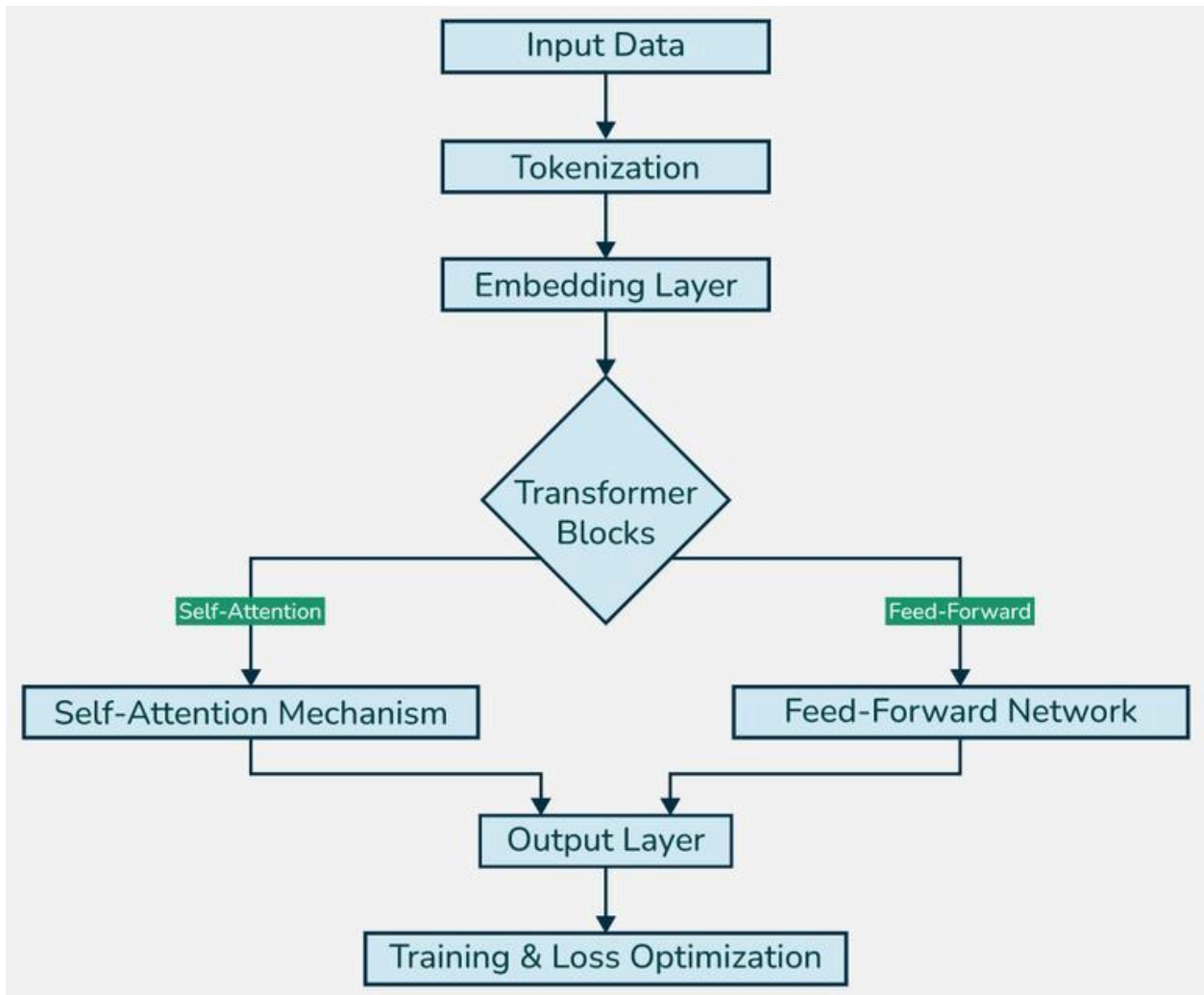


Key, Value



Attention
weights

Query



Training stages of an LLM

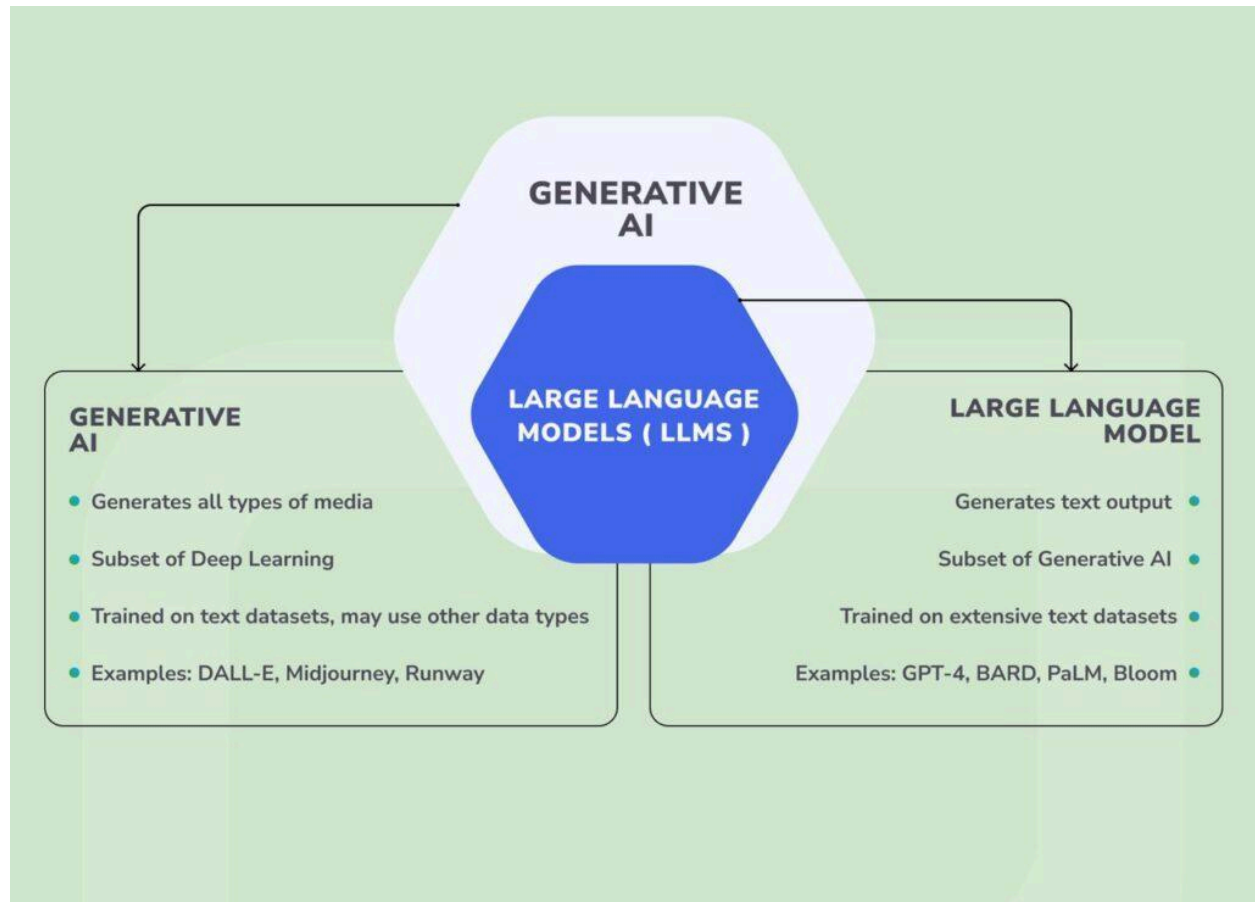
1. Self-supervised learning



2. Supervised learning



3. Reinforcement learning



Learn

- Transformer Architecture
- Self-Attention
- Tokenization
- Embeddings
- Fine-Tuning
- RAG (Retrieval-Augmented Generation)
- Agent Systems

Tools

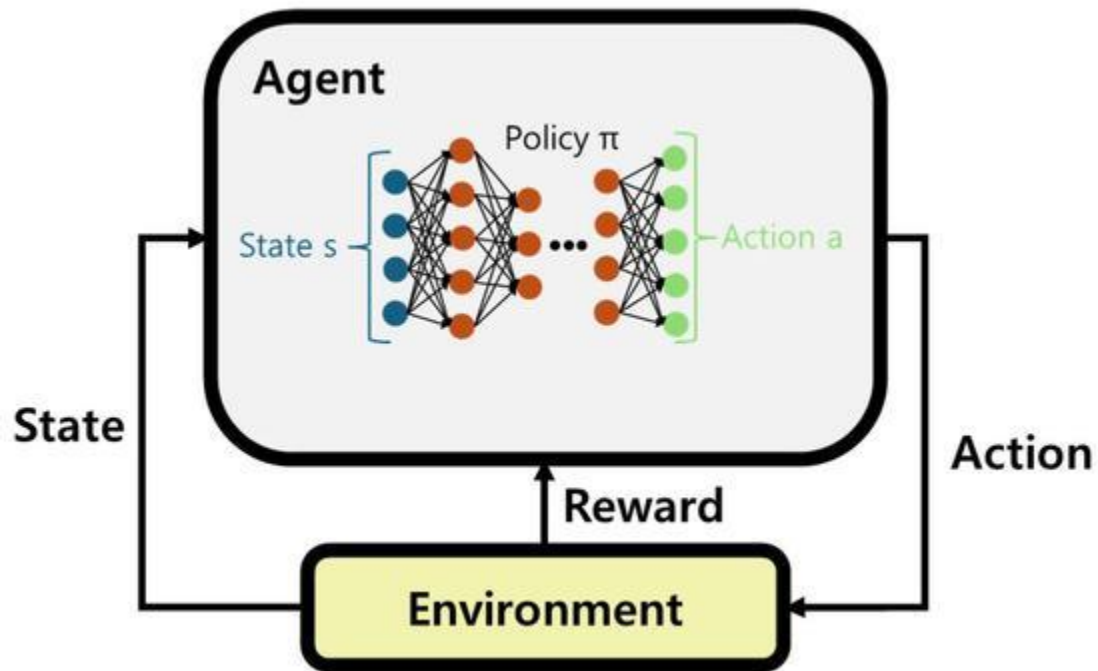
- [Hugging Face Transformers](#)
- [LangChain](#)
- [LlamaIndex](#)

Projects

- Custom Chatbot

- Document Q&A System
 - AI Assistant
 - Local LLM on Raspberry Pi (small models)
-

Phase 5: Reinforcement Learning (RL)



Q-Table

1. Initialize Q-Table and Hypeparamters.

Q-Table						
S1	Up	Up	Down	Left	Right	Right
S1	2	1	0	0	0	0.0
S1	3	4	0	0	0	0.0
S3	4	5	0	0	0	0.0
S9	S1	7	0	0	0	0.1

Feedback

 Reward (R)

 Reward (R)

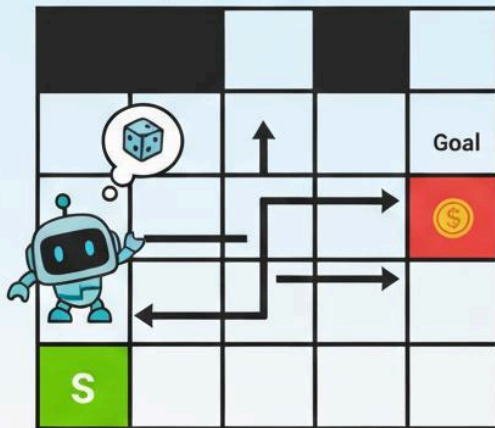
 +10

Next State (S')

Exploration Rate (ϵ)



2. Observe State, Choose Random Action & (Exploration).



$$Q(S_t, A_t) = \frac{\text{Target Value}}{0.1}$$

 0.1

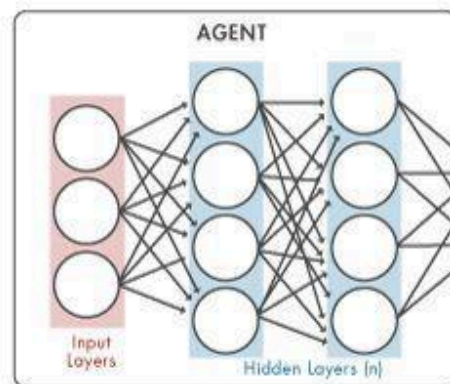
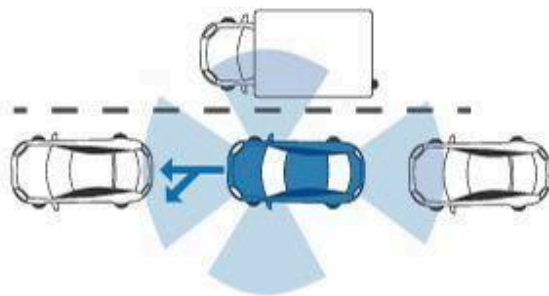
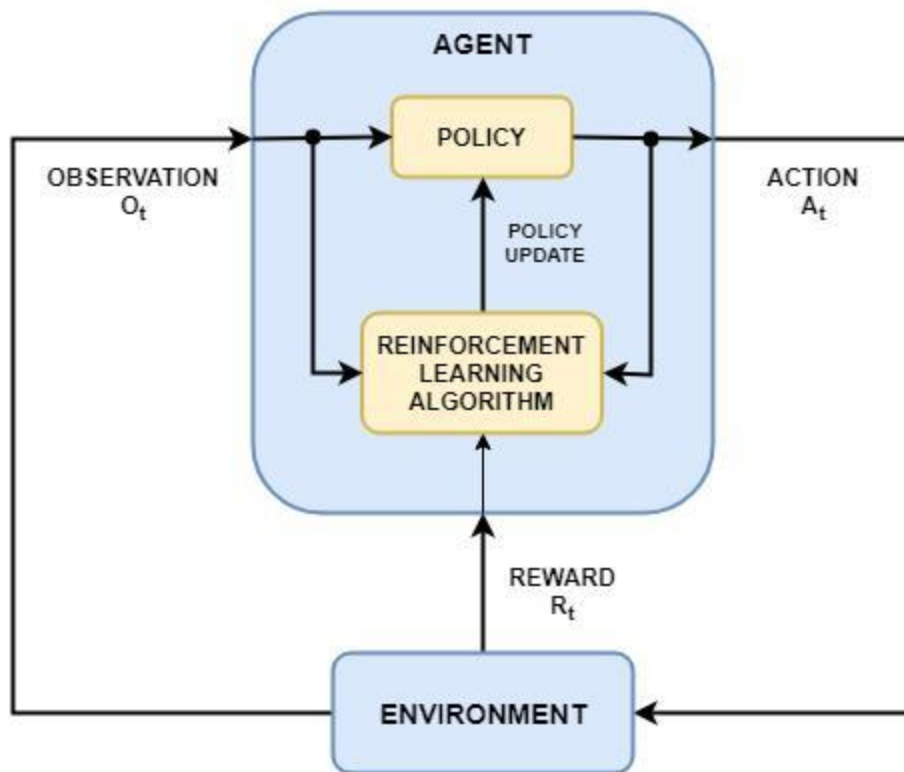
0.1

$$Q(S_t, A_t) + \max Q(a, A)$$

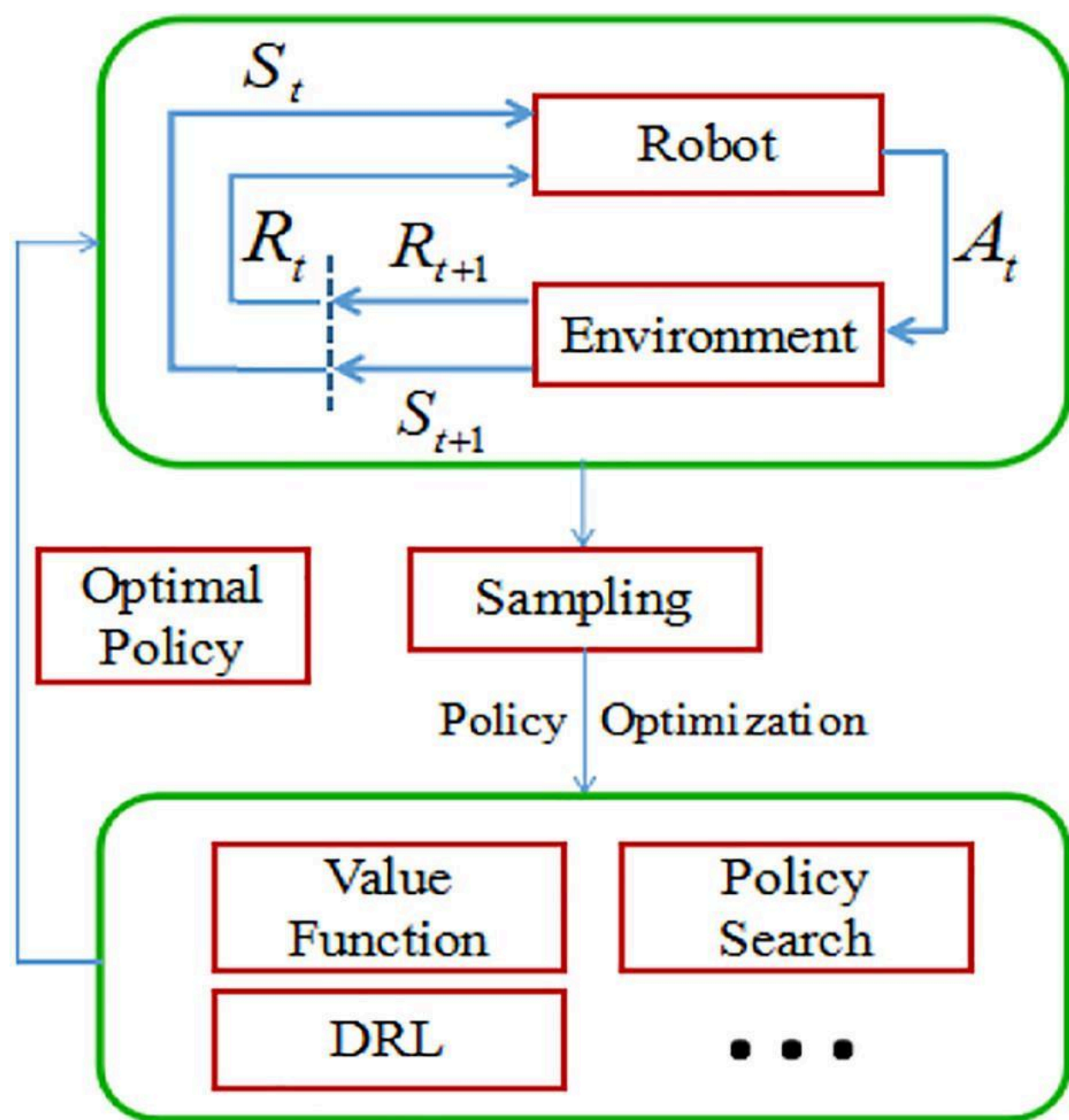
Target Value

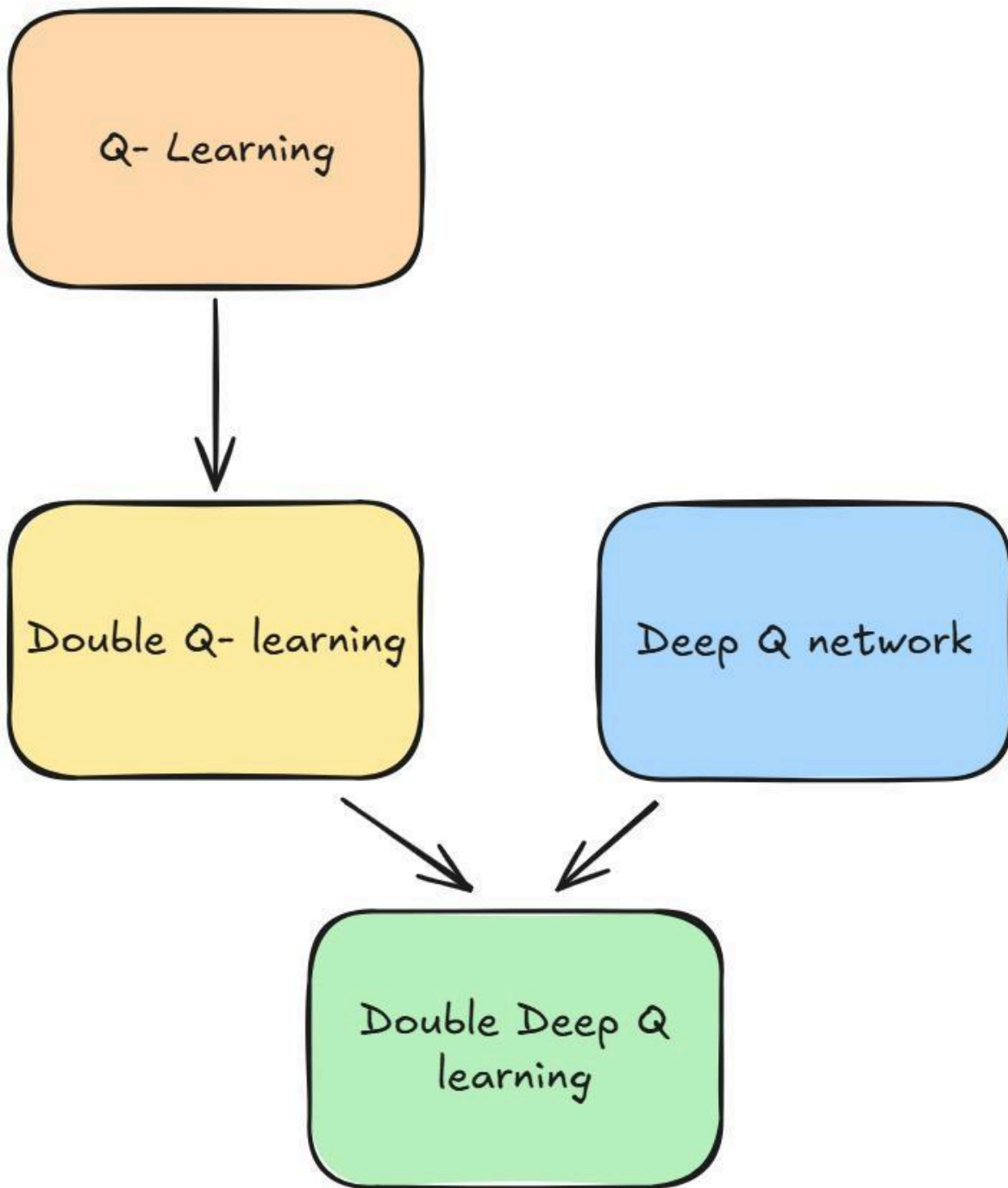
Up	Q _t	0	0	0
 0	0.1	0	 0	$\gamma \max Q(a, A)$
 0	0	 0	 0	$\gamma \max Q(a, A)$

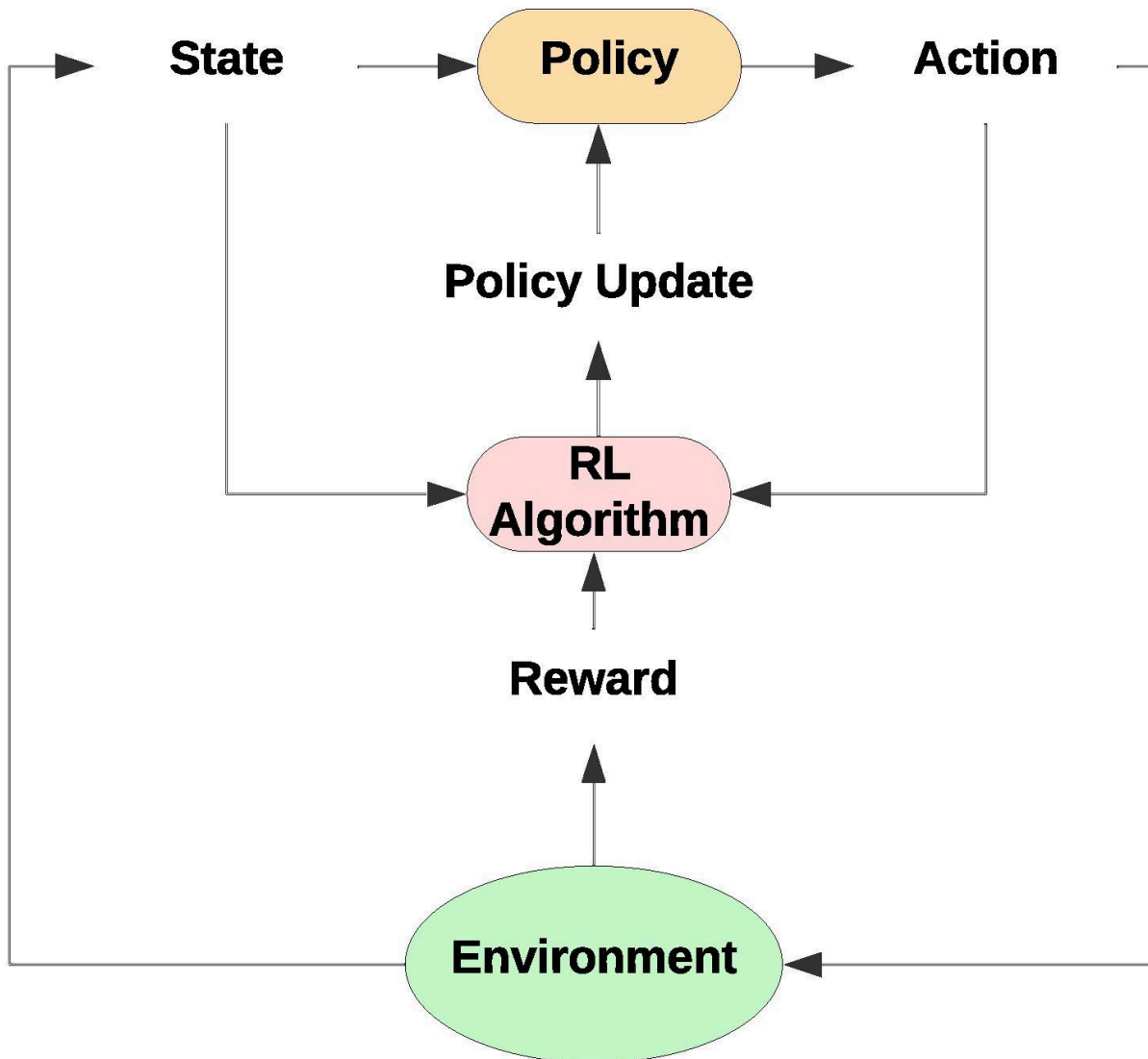
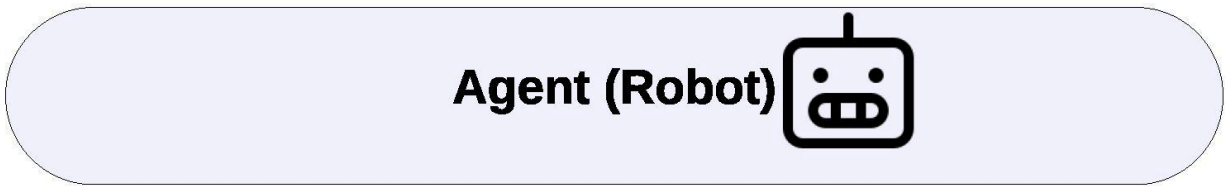
5. Repeat. Q-Values Converge.
Optimal Policy Found!



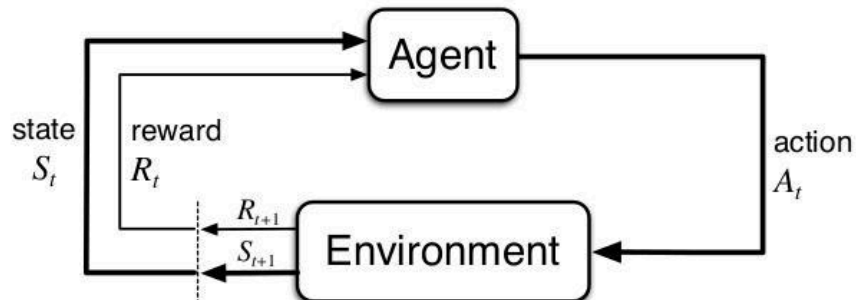
- TURN LEFT
- TURN RIGHT
- BRAKE
- ACCELERATE







The Agent-Environment Interface



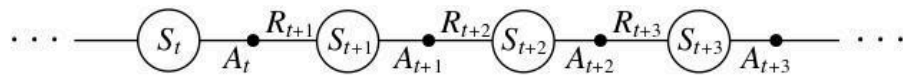
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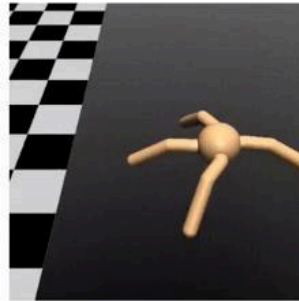


Deep Reinforcement Learning (Deep RL)

Deep Learning



Deep RL



- **What is it?** Framework for learning to solve sequential decision making problems.
- **How?** Trial and error in a world that provides occasional rewards
- **Deep?** Deep RL = RL + Neural Networks

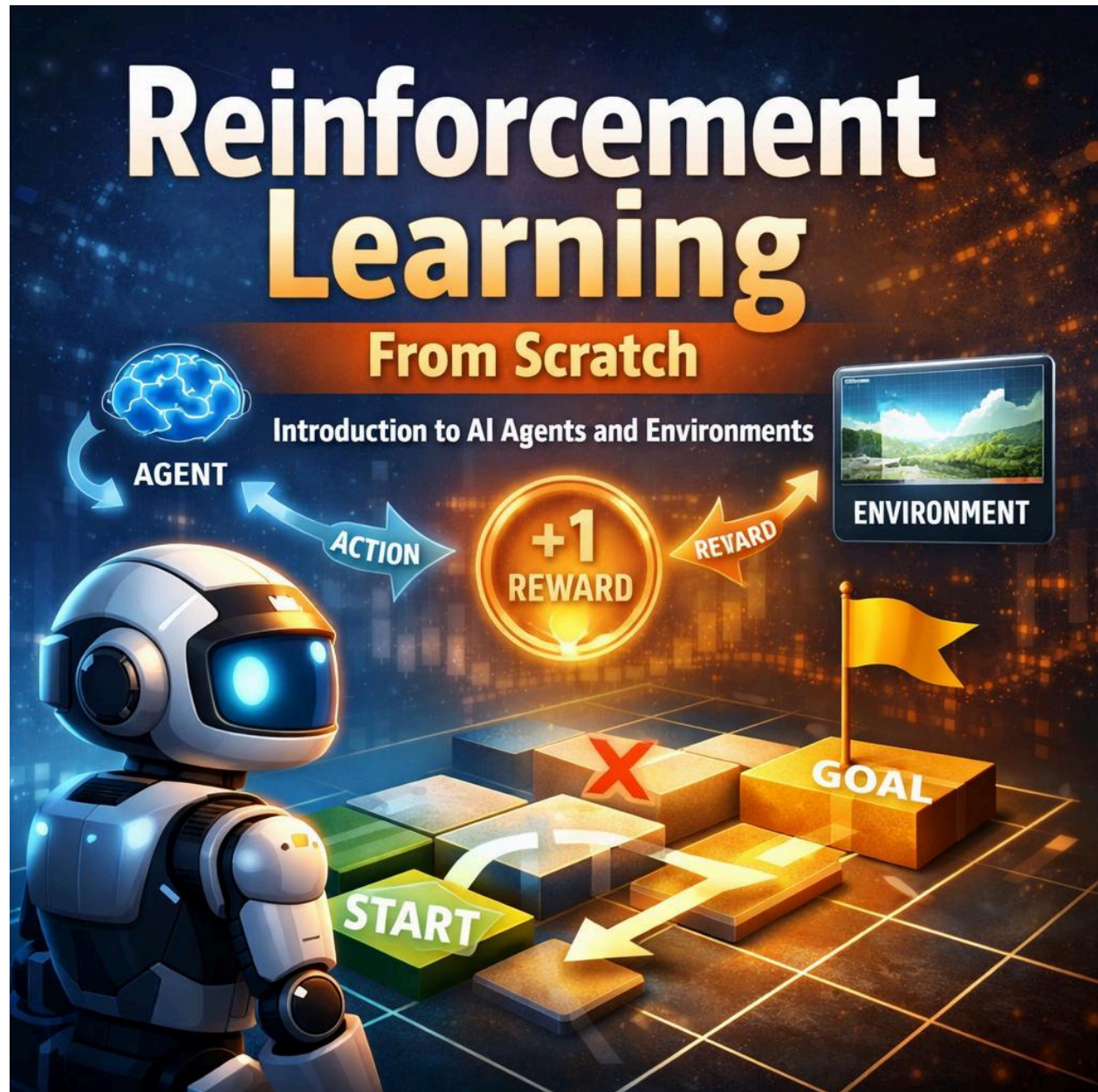


For the full list of references visit: <https://fbcal.mit.edu/references> [306, 307]

<https://deeplearning.mit.edu> 2019

you can reason about it I would take
that power and apply it to





Learn

- Markov Decision Process (MDP)
- Q-Learning
- SARSA
- DQN
- PPO
- A3C
- Multi-Agent RL

Tools

pip install gymnasium stable-baselines3

Projects:

- Self-Learning Game Agent
 - Robot Navigation
 - Drone Path Planning
 - Autonomous Vehicles
-

Phase 6: Research Skills

Read Papers From

- [arXiv](#)
- [Papers With Code](#)

Learn

- Reading Research Papers
 - Experiment Design
 - Benchmarking
 - Model Evaluation
 - Scientific Writing
-

Phase 7: Advanced Research Areas

Computer Vision

- Vision Transformers (ViT)
- YOLO
- Segmentation
- 3D Vision

Generative AI

- Diffusion Models
- GANs
- Multimodal AI

Robotics AI

- SLAM
- ROS2
- Autonomous Navigation

Edge AI

- TensorFlow Lite
 - ONNX
 - Quantization
-

High-Impact Research Projects

1. Multimodal AI Assistant

- Vision + Language
- Voice + Camera
- LLM Integration

2. Thermal Vision AI

Your MLX90640 + Raspberry Pi 5 setup can be used for:

- Human Detection
- Wildlife Monitoring
- Security Systems

3. Autonomous Surveillance Robot

- Raspberry Pi 5
- YOLO
- Thermal Camera
- LoRa Communication

4. AI Drone Research

- Object Tracking
- Navigation
- RL-Based Control

5. Edge AI Network

- ESP32
- Raspberry Pi
- LoRa
- TinyML

Essential Tools

Area	Tools
Programming	Python, C++
Deep Learning	PyTorch, TensorFlow
LLMs	Transformers, LangChain
Computer Vision	OpenCV, YOLO
RL	Gymnasium, Stable-Baselines3
Deployment	Docker, Kubernetes
Research	Jupyter, arXiv
